ILLINOIS
Grainger College of Engineering

ME371
Final Presentation

Team #7 AL2
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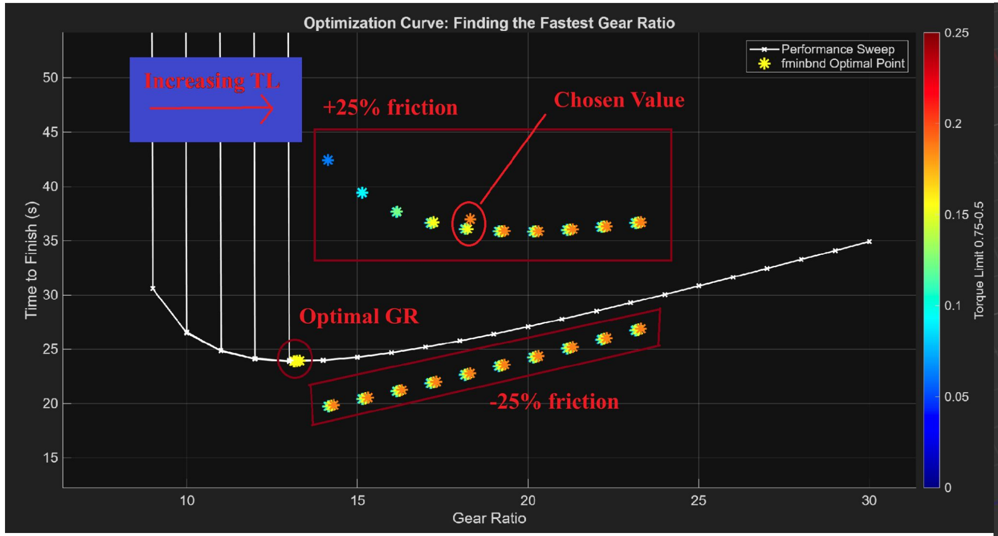
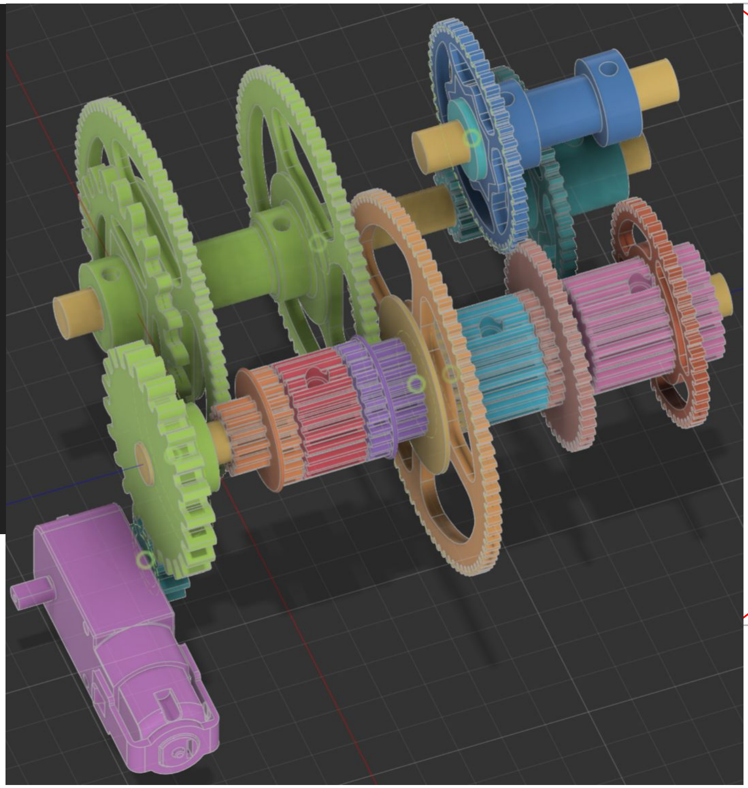
Optimization



Power Transmission



Shifting

	Reverse	4 (high speed)	3	2	1
Ratio	1:1	2:1	1:3	1:12	1:18

Machine Component Analysis

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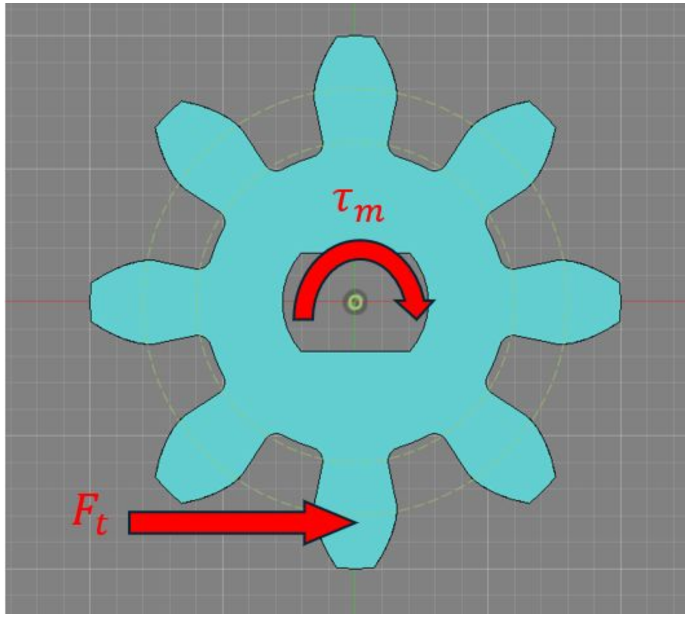


Table 15.1 Overload Correction Factor K_o

	Driven Machinery		
Source of Power	Uniform	Moderate Shock	Heavy Shock
Uniform	1.00	$\boxed{1.25}$	1.75
Light shock	1.25	1.50	2.00
Medium shock	1.50	1.75	2.25

Table 15.2 Mounting Correction Factor K_m

	Face Width (in.)			
Characteristics of Support	0 to 2	6	9	16 up
Accurate mountings, small bearing clearances, minimum deflection, precision gears	1.3	1.4	1.5	1.8
Less rigid mountings, less accurate gears, contact across the full face	$\boxed{1.6}$	1.7	1.8	2.2
Accuracy and mounting such that less than full-face contact exists	Over 2.2			

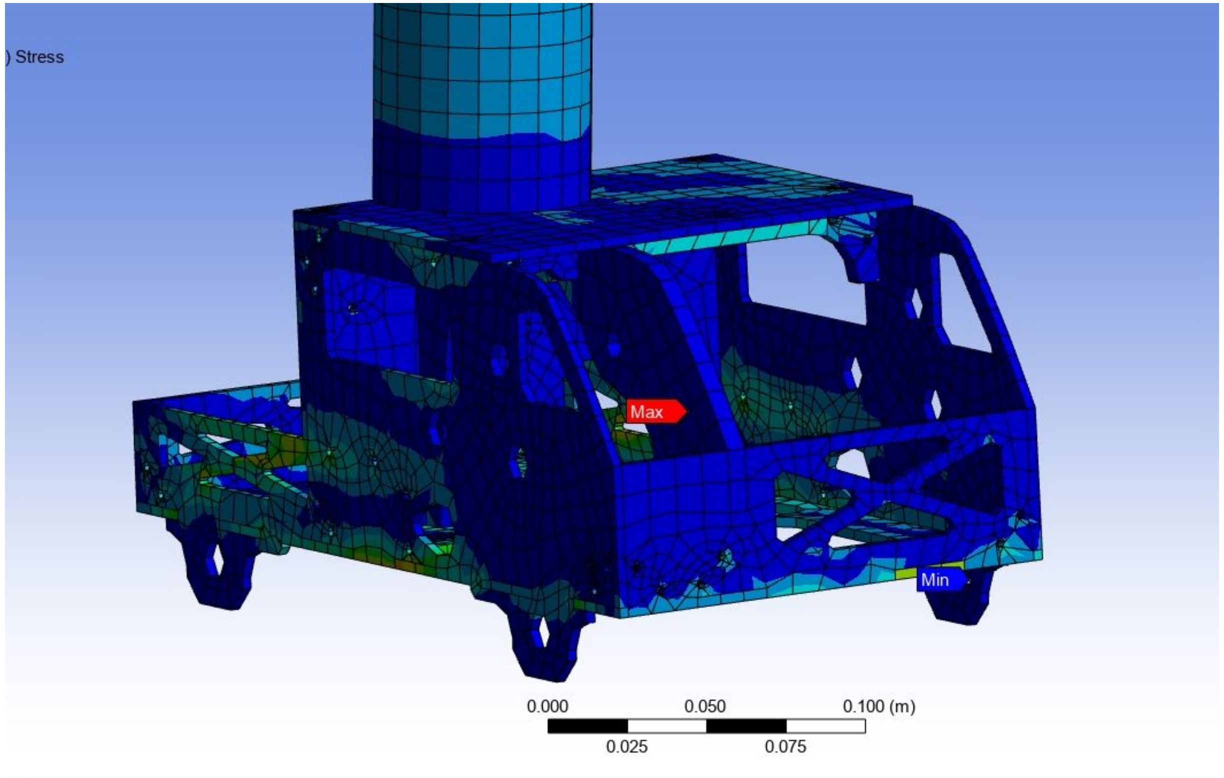
<u>Assumptions:</u>

- Cyclic loading
- Max power output
- Constant rotational speed
- Low-medium quality part

$$F_t = \frac{\tau_m}{d/2} = 4.875\ N$$

$$\sigma = \frac{F_t P}{bJ} K_v K_o K_m = 94.2\ kPa \ll \sigma_u = 131\ MPa$$

FEA Stress Analysis



Component: Chassis

Scenario: Impact

Assumption: Bonded Connections

Software: ANSYS

- Results:
  - σ_{max} = 1.17 MPa
  - δ_{max} = 0.0151 m
  - FS= 26.75
  - Max deflection at right, rear wheel

Summary of Heptathlon Performance

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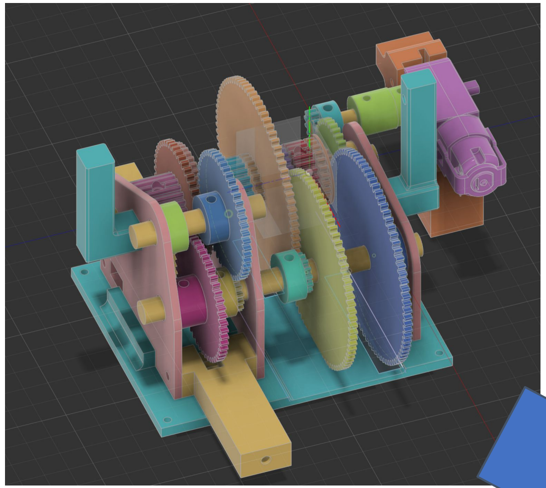
Performance Metric	George’s Need	Performance
Cost (1pt)	<$50	$27.43
Quality (1pt)	High quality with 5-gear capability	4 forward gears + reverse
Speed (2pts)	1.0m/s	1.04m/s
Agility (2pts)	8 1m traversals in 60s	7.5 traversals
Strength (2pts)	Pull S3.0kg up 30° slope	3.0kg
Efficiency (1pt)	40%	93.33%
Durability (1pt)	Survive 3kg dropped from .25m	3kg @ .25m

Overall performance: 98.75%

Transmission:

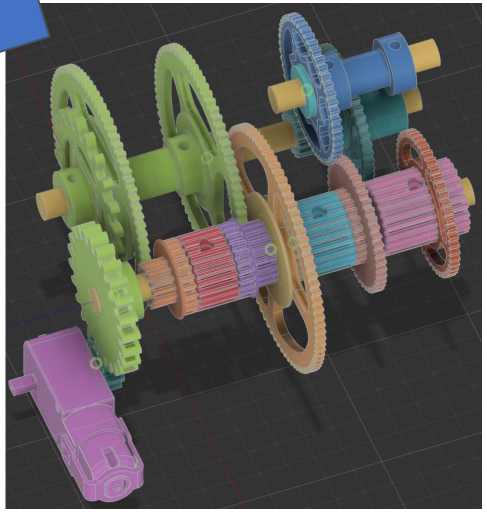


Idea:

- Originally incorporated its own baseplate and walls
- Intended to make more accessible and easier to iterate

Reality:

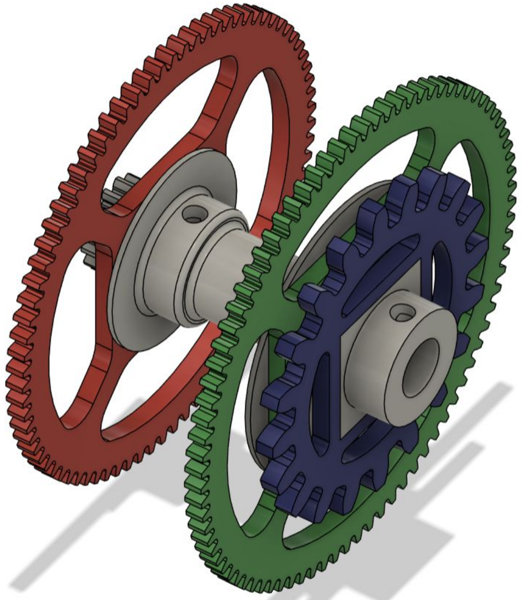
- Made assembly process needlessly long and difficult
- Caused certain components to be less accessible than intended
- Forced chassis to be larger than necessary



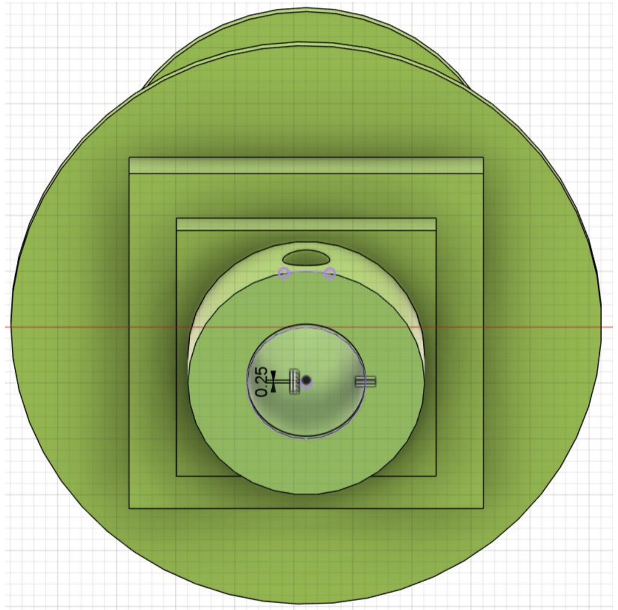
Biggest Challenge: Shifting into 4th gear

Background: Excessive friction and gear misalignment caused the motor to stall when shifting into 4th gear

To fix diagonal misalignment: “Christmas tree” style gears



To fix eccentricity: offset center hole



To reduce friction: delrin inserts inside axle holes